

Farhan Akhtar Makandar

AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Versatile AI & Machine Learning Engineer with comprehensive expertise spanning machine learning, deep learning, generative AI, agentic AI, and AI-powered software development. Research experience at NIT Calicut on transformer architectures for time-series prediction with 22% improvement over traditional baselines. Skilled in building end-to-end AI systems from research prototypes to production deployments. 11 industry certifications from IBM, NVIDIA, AWS, Oracle, and Coursera validating cross-domain AI proficiency.

TECHNICAL SKILLS

- AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Transformers
- GenAI: LLM Fine-Tuning, RAG Pipelines, Prompt Engineering, LangChain, Vector Databases
- Agentic AI: Multi-Agent Systems, Autonomous Agents, Tool Use, Chain-of-Thought
- Deep Learning: CNN, RNN, LSTM, GANs, Attention Mechanisms, SAINT
- Data Science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Jupyter
- Programming: Python, SQL, C, R, HTML/CSS, React, Next.js, TypeScript
- Cloud: AWS, Oracle Cloud, MongoDB, Git/GitHub, Power BI, Vercel

PROFESSIONAL EXPERIENCE

Research Intern - AI/ML - NIT Calicut

Jun 2024 - Dec 2024

- Developed and evaluated transformer-based models for complex predictive tasks using PyTorch, achieving rigorous performance benchmarks
- Collaborated on cutting-edge AI research focused on neural network optimization and deep learning applications for engineering problems
- Conducted comprehensive data analysis, feature engineering, and model validation using scientific computing libraries
- Assisted in documenting research findings and preparing technical reports for publication

PROJECTS

Seismic Response Prediction - TFT

Temporal Fusion Transformer model achieving 22% improvement over XGBoost and KNN for seismic structural response prediction. Full pipeline: EDA, feature engineering, model training, and evaluation.

Techn

Agentic AI System - RAG Pipeline

End-to-end RAG pipeline with vector database, LLM reasoning, and context-aware response generation for intelligent document understanding.

Techn

SAINT Model & Sales Analytics

Attention-based classification model for tabular data and Power BI dashboard with DAX formulas for real-time business intelligence.

Techn

3D Portfolio Website

Production-ready interactive portfolio with 3D particle field, custom GLSL shaders, Framer Motion animations, and contact form backend.

Techn

EDUCATION

B.E. in Artificial Intelligence & Machine Learning

Yenepoya Institute of Technology, Moodbidri (VTU) | Expected 2026

CERTIFICATIONS

- IBM Python for Data Science (Credly Verified)
- Machine Learning with Python - IBM Cognitive Class
- Deep Learning Fundamentals - IBM Cognitive Class
- Accelerating Deep Learning with GPUs - IBM Cognitive Class
- Supervised Machine Learning - Coursera
- Building RAG Agents with LLMs - NVIDIA
- AWS Generative AI Essentials - AWS
- AWS Prompt Engineering - AWS
- Oracle Cloud Infrastructure 2025 AI Foundations Associate
- Accenture Data Analytics Simulation - Forage
- MongoDB Atlas - MongoDB University
- HTML Training - IIT Bombay